Few-Shot Object Detection in Aerial Imagery Guided by Text-Modal Knowledge

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***Abstract*— Few-shot object detection (FSOD) has received numerous attention due to the difficulty and time-consuming of labeling objects. Recent researches achieve excellent performance in a natural scene by only using a few instances of novel classes to fine-tune the last prediction layer of the model well-trained on plentiful base data. However, compared with natural scene objects with a single direction and small size variety, the direction and size of the objects in remote sensing images (RSIs) vary greatly. The methods proposed for the natural scene cannot be directly applied to RSIs. In this article, we first propose a strong baseline for RSIs. It fine-tunes all detector components acting on high-level features and effectively improves the performance of novel classes. Further analyzing the results of the baseline, we find that the error for novel classes is mainly concentrated in classification. It misclassifies novel classes as confusable base classes or backgrounds due to the difficulty in extracting generalized information from limited instances. As is well-known, text-modal knowledge can highly summarize the generalized and unique characteristics of categories. Thus, we introduce text-modal descriptions for each category and propose an FSOD method guided by TExt-MOdal knowledge, called TEMO. Specifically, a text-modal knowledge extractor and a cross-modal assembly module are proposed to extract text features and fuse the text-modal features into visual-modal features. The fused features greatly reduce the classification confusion of novel classes. Furthermore, we introduce a mask strategy and a separation loss to avoid over-fitting and ambiguity of text-modal features. Experimental results on detection in optical remote sensing images (DIOR), Northwestern Polytechnical University (NWPU), and fine-grained object recognition in high-resolution remote sensing imagery (FAIR1M) illustrate that our TEMO achieves state-of-the-art performance in all settings.**

***Index Terms*— Few-shot learning (FSL), object detection, remote sensing images (RSIs), text-modal knowledge.**

Manuscript received 14 December 2022; accepted 24 February 2023. Date of publication 28 February 2023; date of current version 8 March 2023. This work was supported by the National Natural Science Foundation of China under Grant 61725105 and Grant 62171436. *(Corresponding author: Xian Sun.)*

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Digital Object Identifier 10.1109/TGRS.2023.3250448

I. Introduction

BENEFITING from the development of remote sensing and deep learning, significant progress has been made in remote sensing object detection [1], [2], [3]. But almost all deep learning detectors require abundant data for training. Labeling plentiful data is very labor-intensive and time-consuming, especially for remote sensing images (RSIs) with large formats and complex backgrounds. Besides, it is difficult to obtain adequate data for rare categories. In this case, existing detectors will fall into over-fitting. Few-shot object detection (FSOD) is proposed to solve the problem.

FSOD applies the knowledge learned on data-rich base datasets to improve the performance of novel classes with limited data, divided into meta-learning and fine-tuning methods. Meta-learning methods [5], [6], [7] intuitively construct few-shot episodes and extract prototypes from support sets to predict query images. They have to pay extra attention to the transferability across episodes and ignore the learning of discriminative representations of objects [8]. Especially for RSIs, affected by illuminations, loads, object poses, and so on, the differences between episodes are pronounced. Fine-tuning methods [9], [10], [11] transfer the knowledge from base classes to novel classes through fine-tuning. They focus entirely on the representations of categories. Focusing on accurate FSOD, we propose a fine-tuning method for RSIs.

Since natural scene images mostly comprise objects with unitary orientation, small size variations, and are located in the center of images, it is easy to distinguish foreground objects. As shown in Fig. 1(a), a two-stage fine-tuning approach (TFA) [10] achieves remarkable performance by only fine-tuning the last layers of prediction heads on novel classes while freezing all other layers. However, RSIs contain objects with random orientations, dense arrangements, and different sizes [12]. The differences between classes are significant. As stated in FSOD via contrastive proposals encoding (FSCE) [11], it is counter-intuitive that adapt to novel classes without any adjustment of the components acting on high-level features, including feature pyramid network (FPN), region proposal network (RPN), and regions of interest (RoI) feature extractor. It results in the inability to learn semantic information for novel classes. Specifically, FPN cannot accurately map the semantic information of novel classes to the features of all scales, RPN cannot extract abundant proposals containing instances of novel classes, and the RoI feature extractor cannot learn representative information for regression and classification. Thus, we propose a strong baseline in Fig. 1(b). It only freezes the backbone and fine-tunes FPN, RPN, and

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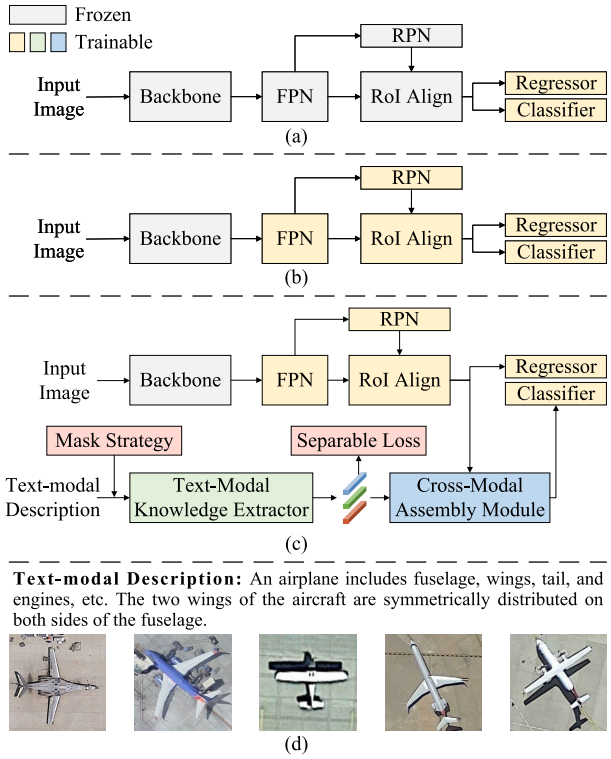


Fig. 1. Schematic of fine-tuning-based FSOD methods and text-modal descriptions. Fine-tuning methods (a) and (b) only extract representations of categories from limited instances. They are unable to extract generalized and unique representations when there are obvious intraclass differences, such as the instances of airplane in (d). However, as shown in (d), text–modal descriptions can summarize representative characteristics of categories in several simple sentences. Hence, our proposed method (c) introduces text-modal knowledge to compensate for the insufficient generalized information extracted from limited instances. (a) TFA. (b) Our baseline. (c) Our method. (d) Text-modal description.

RoI align together with prediction heads. The experimental results demonstrate that the baseline effectively improves the performance of novel classes while alleviating over-fitting.

Further analyzing the performance of the proposed baseline, we find that the error for novel classes is mainly in the classification branch, not the regression branch. The class-agnostic RPN efficiently screens out foreground proposals belonging to novel classes, but the model misclassifies novel classes as confusable base classes or background, as shown in Fig. 2(a) and (b). On the one hand, the objects in RSIs have large intraclass differences, as shown in Fig. 2(c). When fine-tuned with a few instances, the model cannot learn the generalized characteristics of each category. On the other hand, the objects in RSIs also have large interclass similarities. For fine-grained object detection in Fig. 2(d), the differences between different types of airplanes are imperceptible. The fine-tuned model cannot extract unique features for each class with limited instances. As a result, the classifications are confounded, resulting in degraded performance. Similar to FSCE [11], we make a calculation of manually correcting misclassified yet accurately localized box predictions that can increase novel class average precision by about 25 points.

Due to the above-mentioned characteristics of RSIs, the model cannot effectively extract discriminative information of categories when training data are limited. However, text-modal descriptions are highly general and summative, as shown in Fig. 1(d) and Fig. 2(c) and (d). We believe that artificially describing unique and generalized characteristics of categories and introducing these text-modal sentences can assist the model in summarizing the representative information of categories and improve classification accuracy. Since the model only receives RGB images, the text-modal descriptions should mainly contain the characteristics that can be perceived from images, such as appearance, material, structure, surroundings, and so on. Preferably, these characteristics are generic to objects of the same category and distinct from other categories. Although the descriptions require extra annotation, it is cost-effective to only annotate several sentences for each category. Besides, this article is devoted to exploring the feasibility of introducing text-modal knowledge to improve FSOD performance in RSIs, the approaches and qualities of generating descriptions will be researched in the future. Thus, multiple experts in the field of remote sensing are invited to artificially summarize the descriptions of categories in three datasets [2], [3], [13] for subsequent research, as shown in the Appendix.

Based on the summarized text-modal descriptions, we propose an FSOD method for RSIs guided by TExt-MOdal knowledge, named TEMO.[1] As shown in Fig. 1(c), it introduces text-modal knowledge to make up for classification confusion caused by the insufficient representative information of limited data. First, considering the temporal information in descriptions, a text-modal knowledge extractor is proposed to capture long-term dependencies and extract semantic information from text descriptions. Then, since visual-modal features and text-modal features are distributed in different feature spaces and cannot be directly integrated, we propose a cross-modal assembly module to fuse text-modal information into RoI visual features. It supplements the generalized information of categories in visual-modal features. The fused features are sent to the classifier to reduce classification confusion. In addition, we introduce a mask strategy and a separation loss for training the model to extract useful text-modal information from descriptions. The former performs random wordwise masking on descriptions during training to prevent over-fitting from consistently using the same descriptions for each category. The latter improves the discriminability of text-modal information between different categories by constraining text-modal features of each category to be orthogonal to each other. In order to learn to extract useful information, the text-modal descriptions are introduced when both training on base data and fine-tuning on limited novel data. We validate the effectiveness of our proposed method on two general remote sensing object detection datasets detection in optical remote sensing images (DIOR) [2], Northwestern Polytechnical University (NWPU) very-high-resolution (VHR)-10 [13], and a fine-grained object detection dataset fine-grained object recognition in high-resolution remote sensing imagery (FAIR1M) [3]. Experimental results illustrate that our TEMO achieves state-of-the-art performance in all settings.

Our contributions are summarized as follows.

1) We analyze the impact of the differences between nature scene images and RSIs on FSOD and propose a strong baseline based on TFA. It enables the detector to extract

[1]Code will be available soon.

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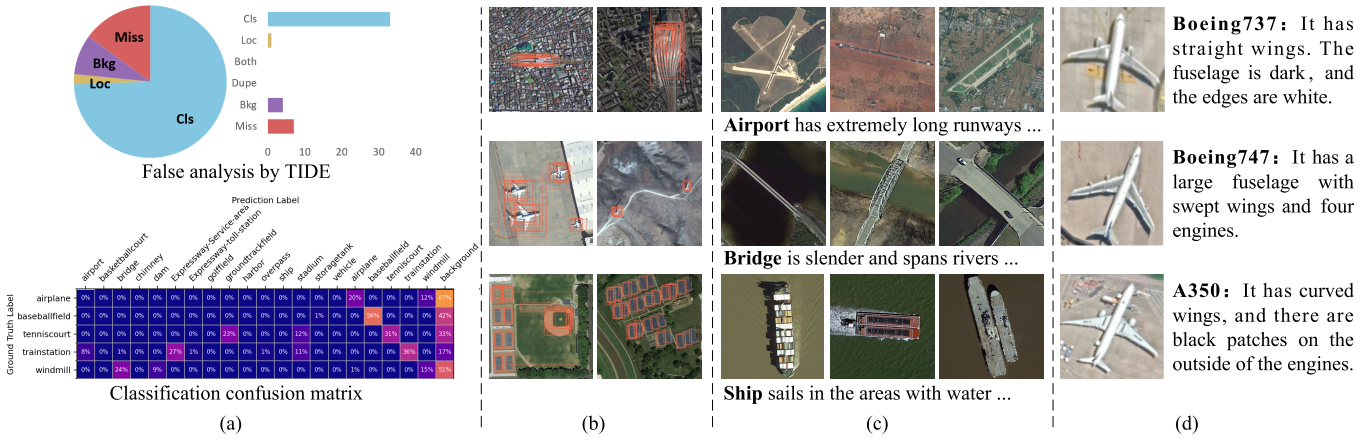


Fig. 2. Analysis of our proposed baseline. (a) Error analysis on DIOR dataset with 10-shot setting. By using toolbox for identifying detection errors (TIDE) toolbox [4] and calculating confusion matrix, the main error is classification error that misclassifies novel classes as confusable base classes or backgrounds. (b) Proposals predicted by RPN for novel classes. RPN efficiently extracts proposals of novel classes. The main reason for classification confusion is that the objects in RSIs have characteristics of large (c) intraclass differences and (d) interclass similarities. It can be seen from (c) and (d) that text-modal descriptions can highly summarize generalized and unique characteristics of categories, so introducing text-modal knowledge will reduce classification confusion and improve the performance of novel classes.

semantic information of novel classes and effectively improves the FSOD performance of RSIs.

2) To alleviate classification confusion in our baseline caused by insufficient information, we propose a novel method to extract text-modal knowledge from human-summarized description sentences and complement generalized and unique information of categories into visual features.
3) We propose a text-modal knowledge extractor to capture long-term dependencies from description sentences with different lengths and generate text-modal features for each category. Besides, a cross-modal assembly module is proposed to fuse text-modal features into visual-modal features via key-value embedding, which enhances generalized information of categories in visual-modal features.
4) A mask strategy is designed to randomly occlude part of words in description sentences during training. It prevents over-fitting from consistently extracting text-modal knowledge from the same descriptions. We also introduce a separation loss to constrain text-modal features of different categories to be orthogonal. It prompts text-modal features to highlight unique visual features and improves distinctiveness between different categories.

II. Related Works

In this section, we review some relevant works on object detection, few-shot learning (FSL), FSOD, and text-modal knowledge used in vision tasks, including progress in the field of computer vision and remote sensing.

A. Object Detection

With the development of convolutional neural networks (CNNs), object detection has made great advances [14], [15], [16], [17], [18]. Popular detectors are divided into two-stage models and single-stage models. Two-stage methods [14], [15] extract candidate proposals and then perform fine classification and regression. Single-stage methods [16], [17], [18] discard proposal extraction and directly use a convolutional network to predict categories and locations. Recently, some transformer-based detectors [19], [20] also achieve comparable performance.

Due to the characteristics of multiscale, dense arrangement, and arbitrary orientation in RSIs, some object detection methods for RSIs have been proposed [21], [22], [23], [24]. R3Det [22] detects rotating objects with a large aspect ratio and dense arrangement. AOPDet [23] introduces a nonsequential corners representation for RSIs. Zhang et al. [24] propose a dynamic knowledge distillation framework to achieve efficient and accurate detectors. LR-TSDet [25] obtains global and multihierarchical features for detecting tiny objects.

Although achieving remarkable performance, these models all require abundant annotated data. Annotating RSIs is more time-consuming and labor-intensive than natural scenes [26], [27]. Besides, for difficult Synthetic Aperture Radar (SAR) images [28], [29], labeling objects requires the participation of professional interpreters, which undoubtedly increases the difficulty of image annotation. Thus, we propose an FSOD method to alleviate the dependence on the number of annotating images.

B. Few-Shot Learning

FSL aims to apply limited data to obtain a model with excellent performance, consisting of augmentation, metric learning, and meta-learning-based methods. Augmentation-based methods [30], [31] enrich limited data by data augmentations. Metric learning-based methods [32], [33], [34] determine the nearest neighbors as prediction results. Meta-learning-based methods [35], [36], [37] construct episodes and learn knowledge from annotated support sets to predict query sets.

Recently, some works have been presented for RSIs [38], [39]. Li et al. [40] adopt attention to discover discriminative features. RS-MetaNet [41] combines meta-learning and metric learning for classification. SPNet [42] proposes a siamese-prototype network to increase the validity of prototypes.

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These FSL methods only focus on image classification and cannot be directly applied to object detection where the positions of objects need to be predicted.

C. Few-Shot Object Detection

FSOD has gradually attracted the attention of researchers. It learns from a data-rich base dataset and a novel dataset with limited data to improve the performance of novel classes. It includes data augmentation methods, meta-learning methods, and fine-tuning methods. Data augmentation methods [43], [44] improve data diversity through data augmentation or feature transformation. Meta-learning methods [5], [6], [7], [45], [46], [47] construct episodes, including support sets and query sets, and train models to extract information from support sets for detecting objects in query sets. Fine-tuning methods [9], [10], [11], [48] transfer the knowledge learned from the base dataset to a novel dataset through fine-tuning models to be equipped to detect novel objects.

Recently, some FSOD methods for RSIs have been presented [49], [50], [51]. Li et al. [12] introduce multiresolution features based on Meta you-only-look-once (YOLO) [5] for detecting multiscale objects. Le Jeune et al. [52] propose a prototypical network based on representation learning. Path-aggregation multiscale few-shot detector (PAMS-Det) [49] applies the involution operator and shape bias to enhance the classification branch and proposes a multiscale path-aggregation module to improve the regression branch. Double head few-shot detection (DH-FSDet) [53] presents a double-head predictor and an annotation sampling and preprocessing strategy. Prototype-CNN [54] converts support images into class-aware prototypes and injects these prototypes into RPN to guide the generation of proposals.

These FSOD methods only use limited annotated images to improve the performance of novel classes. However, the insufficient images always lack the generalized and unique features of each class. Our proposed method introduces text-modal knowledge to supplement the respective semantic information of each category to the visual-modal features.

D. Text-Modal Knowledge in Vision Tasks

Text-modal knowledge is often introduced in vision tasks such as image captioning, text–image retrieval, and zero-shot learning. Image captioning [55], [56] and text–image retrieval [57], [58], [59] methods learn semantic and visual embeddings and map them to the same space for image-text matching. Zero-shot learning methods [60], [61], [62] use different forms of text-modal knowledge to describe the attributes of categories, such as using category names [63] or vectors composed of certain attributes [64] as prior semantic knowledge. They learn the mapping between visual and semantic space to compensate for the lack of visual representation of unseen classes. A recent approach [65] calls on multiple domain experts from the remote sensing field to depict each scene category by one summarized sentence after checking over ten random image scenes from one given category.

Text-modal knowledge can supplement the generalized information of categories missing from limited images, so introducing text-modal knowledge into FSOD can improve the performance of detecting novel classes. Semantic relation reasoning (SRR)-FSD [66] uses a dynamic relation graph network to interact with the embedding of category words and projects visual representations of objects into the embedding space. However, it only uses the words of categories and does not introduce text-modal descriptions. Thus, it cannot provide sufficient semantic generalized information of categories for enriching visual features. Our method introduces text-modal sentences describing the properties of each category and equips text-modal knowledge with visual features for enhancing representative information of categories.

III. Methodology

In this section, we first briefly describe TFA [10] and our proposed baseline. Next, the overall architecture of our TEMO is elaborated. Then, we separately present the designed components, including the text-modal knowledge extractor, cross-modal assembly module, mask strategy, and separation loss.

A. TFA and Strong Baseline

FSOD aims to train a detector on a base dataset $\mathcal{D}_b$ with abundant data for base classes C_b and a novel dataset $\mathcal{D}_n$ containing limited data for novel classes C_n (where $C_n \cap C_b = \oslash$) to improve the performance on novel classes. $\mathcal{D}_b$ contains adequate data (x_i, y_i), where x_i is the image containing the objects belonging to C_b, and y_i represents the category and bounding box position of objects. $\mathcal{D}_n$ only contains a few annotated images with objects belonging to C_n. For k-shot, there are k instances for each category in $\mathcal{D}_n$. FSOD applies limited images in $\mathcal{D}_n$ to quickly transfer the knowledge learned from $\mathcal{D}_b$ into $\mathcal{D}_n$ to improve the performance of C_n.

As a fine-tuning-based FSOD method, TFA [10] applies a two-stage training strategy with base model training and few-shot fine-tuning stages. In the base model training stage, the detector is trained on the abundant data of $\mathcal{D}_b$ to detect objects belonging to C_b. In the few-shot fine-tuning stage, TFA introduces an instance-level feature normalization and a cosine similarity classifier. It creates a balanced training set consisting of $\mathcal{D}_n$ and a subset of $\mathcal{D}_b$, which contains k objects for each category. And the last prediction layers for novel classes are randomly initialized. Then, as Fig. 1(a) shows, only the last layers of the classification and regression heads are fine-tuned on the created training set while keeping the other layers fixed. TFA proves that for the detector well-trained on $\mathcal{D}_b$, only fine-tuning the last prediction layer on $\mathcal{D}_n$ and a subset of $\mathcal{D}_b$ can improve the performance of novel classes.

However, as described in FSCE [11], it is counter-intuitive that directly apply the semantic information learned from C_b into C_n without adjustment. Especially for RSIs, the characteristics between different categories vary greatly. The modules, such as FPN, RPN, and RoI feature extractor, all act on high-level features extracted by the backbone. If these modules are not fine-tuned on $\mathcal{D}_n$, FPN will not accurately map high-level semantic information of C_n into features through a top-down architecture with lateral connections. RPN will

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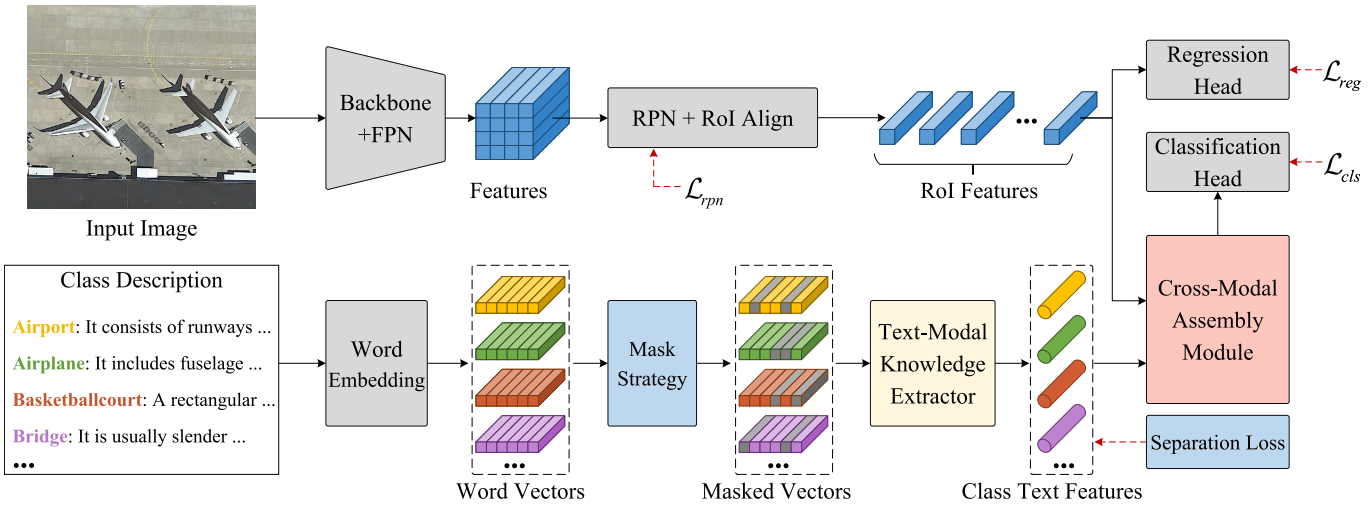


Fig. 3. Architecture of TEMO. It consists of a visual detection branch (top) and a text knowledge extraction branch (bottom). Based on faster RCNN, the visual detection branch extracts RoI features that may contain objects from the input image. And the RoI features are directly used for location regression. The text knowledge extraction branch introduces text-modal descriptions of each category and extracts generalized class text features which are fused into RoI visual features, so as to highlight the category-related information of RoI features and send them to the classification head for reducing misclassification.

tend to predict abundant proposals containing objects of C_b and select a few proposals containing objects of C_n. RoI feature extractor will not learn and retain the representative characteristics of C_n, so reducing the regression and classification accuracy of C_n. Thus, when applying the information about C_b to C_n directly, it is prone to reduce the performance for detecting novel classes due to bringing a lot of noise. In addition, the backbone contains a large number of training parameters. If the backbone is fine-tuned with limited novel data, it will face over-fitting. Thus, it is necessary to freeze the backbone to prevent over-fitting, and by fine-tuning subsequent modules, the high-level features extracted from the frozen backbone are adapted to detect objects belonging to C_n.

To learn the semantic information of novel classes effectively, as shown in Fig. 1(b), we build a strong baseline for RSIs based on TFA. In the first base model training stage, adopting the same training strategy as TFA, the proposed baseline uses abundant images in $\mathcal{D}_b$ to train a detector that performs well on C_b. In the second few-shot fine-tuning stage, different from TFA, the proposed baseline unfreezes FPN, RPN, and RoI align. It fine-tunes them together with the prediction heads on $\mathcal{D}_n$ and a subset of $\mathcal{D}_b$ while only fixing the parameters of the backbone. Compared with TFA, our proposed baseline adjusts the semantic information learned from C_b to fit the characteristics of C_n. Experimental results of freezing different layers in the few-shot fine-tuning stage demonstrate that the proposed baseline greatly improves the performance of novel classes compared with the original TFA.

B. Overview of TEMO

By analyzing the results of our proposed baseline, we found that the error of novel classes is mainly in the classification branch. It misclassifies the novel classes as base classes or backgrounds because the limited instances make it impossible to extract generalized and unique features for each category. Encouragingly, text-modal descriptions can provide a summary of the characteristics of each category. We invite experts in the field of remote sensing to artificially summarize the characteristics of categories in text-modal sentences for both base and novel classes. These sentences summarize the common characteristics contained in each category. See the Appendix for details. Even if the characteristics of each category are obvious, the sentences summarized by each expert may be different. Therefore, the robustness of different description sentences is important for the model. In the end-to-end training, our model learns adaptively based on the support images and input sentences. For different sentences, as long as the characteristics of each category are accurately described without adding noise information, the model can learn to extract similar semantic information from text-modal sentences and map visual-modal features and text-modal features to a stable feature space, ensuring the stability of the performance under different descriptions.

Thus, we propose an FSOD method guided by text-modal knowledge, named TEMO. It introduces text-modal knowledge on the basis of the proposed baseline to supplement the lack of generalized and unique semantic information in visual-modal features. As shown in Fig. 3, our proposed TEMO mainly consists of a visual detection branch and a text knowledge extraction branch. The visual detection branch adopts a two-stage detector faster regions with convolutional neural network (RCNN) [14], including backbone, FPN, RPN, RoI align layer, and prediction heads. Given an input image, the backbone and FPN extract multiscale features which are fed into RPN and RoI align layer to generate RoI features. The regression head predicts the location of objects based on RoI features directly. For the purpose of reducing the classification confusion of novel classes, the classification head classifies objects by using features that combine RoI visual-modal features and text-modal features.

The text knowledge extraction branch consists of a word embedding layer, text-modal knowledge extractor, cross-modal assembly module, mask strategy, and separation loss. First,

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Algorithm 1 Training Process of TEMO

Input:

(x, y): images and labels in $\mathcal{D}_b$ and $\mathcal{D}_n$

S: text-modal description sentences for each category

lr: learning rate

Output:

θ: The weights of the detector that performs well on novel classes

for **stage** in [base training, few-shot fine-tuning] **do**
  Forward propagating the image x to the visual detection branch to extract RoI features f_{vis}
  Send f_{vis} to the regression head to predict location
  Wordwise masking of S using the mask strategy
  Forward propagating the masked S to the text knowledge extraction branch to extract class text features f_{text}
  Forward propagating f_{vis} and f_{text} to the cross-modal assembly module to generate fused features
  Send fused features to the classification head to predict categories
  Calculate the separation loss $\mathcal{L}_{sepa}$ of f_{text}
  Calculate the detection loss $\mathcal{L}_{rpn}$, $\mathcal{L}_{reg}$, and $\mathcal{L}_{cls}$
  Get the total training loss $\mathcal{L}$ in Eq. (1)
  Backward propagation and update $\theta \leftarrow \theta - lr\frac{\partial \mathcal{L}}{\partial \theta}$
end for

the text-modal sentences describing the characteristics of each class are converted into word vectors through the word embedding layer. Then, the text-modal knowledge extractor extracts class text features of each category from the word vectors of the corresponding category. Next, the cross-modal assembly module fuses RoI visual-modal features and text-modal features to enhance the semantic information of each category in RoIs. Finally, the fused features are sent into the classification head for prediction. By introducing description sentences that highly generalize the characteristics of categories, the text knowledge extraction branch generates summative and generalized text-modal features for each category and integrates them into RoI visual features to emphasize the discriminating information and reduce the misclassification of the classifier.

Additionally, we introduce a mask strategy and a separation loss $\mathcal{L}_{\text{sepa}}$ to train the text knowledge extraction branch efficiently. The former randomly generates masking vectors and masks word vectors to avoid over-fitting from consistently training on the same text-modal description sentences. The latter focuses on the separability of class text features between different categories. Therefore, the total training loss of our TEMO is as follows:

$$\mathcal{L} = \mathcal{L}_{\text{rpn}} + \mathcal{L}_{\text{reg}} + \mathcal{L}_{\text{cls}} + \lambda\mathcal{L}_{\text{sepa}} \tag{1}$$

where $\mathcal{L}_{\text{rpn}}$ represents the classification and regression loss of RPN. $\mathcal{L}_{\text{reg}}$ and $\mathcal{L}_{\text{cls}}$ denote regression loss and classification loss of the prediction head, respectively. The regression loss uses L1 loss, and the classification loss applies cross-entropy loss. λ is a balance factor used to control the impact of $\mathcal{L}_{\text{sepa}}$. The training process of TEMO is shown in Algorithm 1.

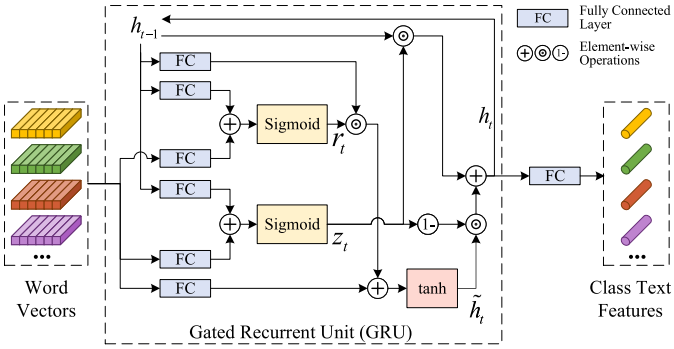


Fig. 4. Illustration of text-modal knowledge extractor. It extracts class text features for each category from the word vectors of the corresponding description sentences.

C. Text-Modal Knowledge Extractor

Text-modal description sentences are able to summarize salient characteristics of categories clearly and concisely, but the lengths of sentences for different categories may be different. In order to extract useful information from these text-modal sentences of different categories, it is necessary to pad the sentences of each category to the same length and convert the words in sentences into word vectors through the word embedding layer.

Suppose the text-modal description of class c consists of several sentences S_c with T_c words $\{w_t\}_{t=1}^{T_c}$. First, the sentences of each class are padded to the same length T, where $T = \max\{T_1, T_2, \ldots, T_C\}$, and C is the number of classes. Then, through a word embedding layer W_e initialized by skip-thought model [67], each word is converted into a d_w-dimensional vector v_t, as shown in the following:

$$v_t = W_e(w_t), \quad t \in [1, T]. \tag{2}$$

To extract class text features from the word vectors of each category, we propose a text-modal knowledge extractor. Considering the temporal information in sentences, the extractor should have a strong ability of long-term memory to learn the comprehensive information contained in the entire descriptions. The same as the skip-thought model, taking into account both the long-term memory ability and the amount of computation, we apply a gated recurrent unit (GRU [68]) containing a reset gate r_t and an update gate z_t to model the class text features of each class from the corresponding word vectors, as Fig. 4 shows. The update formulas are as follows:

$$r_t = \sigma(W_r v_t + U_r h_{t-1}) \tag{3}$$

$$z_t = \sigma(W_z v_t + U_z h_{t-1}) \tag{4}$$

$$\widetilde{h}_t = \tanh(W_h v_t + r_t \odot U_h h_{t-1}) \tag{5}$$

$$h_t = z_t \odot h_{t-1} + (1 - z_t) \odot \widetilde{h}_t \tag{6}$$

where $W_{[\cdot]} \in \mathbb{R}^{d_s \times d_w}$ and $U_{[\cdot]} \in \mathbb{R}^{d_s \times d_s}$ represent fully connected layers. d_s is the dimension of sentence vectors. σ and tanh mean sigmoid and tanh activation functions, respectively. $\odot$ is the elementwise multiplication. h_{t-1} is the final state at the previous moment. $\widetilde{h}_t$ and h_t represent the temporary state and the final state at the current moment. For the sentences with T words, the information of all words in the sentences is united after the T-step update of GRU. And a d_s-dimensional

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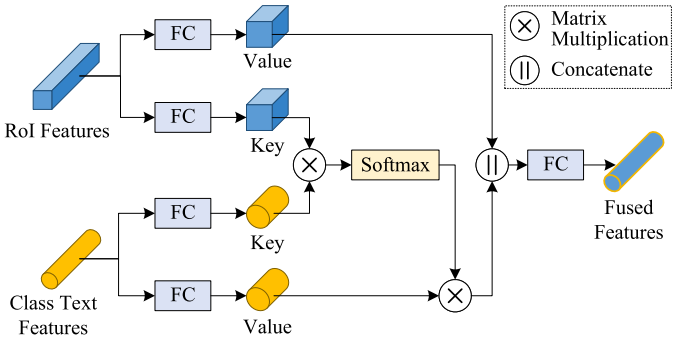


Fig. 5. Illustration of cross-modal assembly module. It integrates RoI visual features and class text features to emphasize the category-related visual features in RoI features.

sentence vector h_T is obtained for each category, which is then sent to a fully connected layer $W \in \mathbb{R}^{d \times d_s}$ to generate the final class text features $f_{\text{text}} = Wh_T \in \mathbb{R}^d$.

The above-mentioned steps are repeated for each category to extract class text features from the word vectors of the corresponding text-modal sentences. The generated class text features f_{text} fully draw on the information contained in text-modal sentences and represent the unique and generalized characteristics of categories, which are subsequently used to enhance RoI visual-modal features to alleviate classification confusion.

D. Cross-Modal Assembly Module

The RoI visual-modal features extracted by the visual detection branch contain visual details of the objects in proposals. And the class text features generated by the text knowledge extraction branch contain the specific and generalized characteristics of categories. Fusing the features of the two modalities can match visual features with the characteristics of each category and enhance category-related information in the visual-modal features to reduce classification confusion.

Since the extracted visual-modal features and text-modal features are in their own feature spaces, directly merging them will generate features with numerous noises due to information mismatch, which drowns out the introduced prior knowledge. Inspired by the key-value embedding of images in dense relation distillation with context-aware aggregation network (DCNet) [46], we propose a cross-modal assembly module to integrate text-modal knowledge into RoI visual-modal features effectively and comprehensively. As shown in Fig. 5, given the RoI visual features $f_{\text{vis}} \in \mathbb{R}^d$ and the d-dimensional class text features f_{text} generated by the text-modal knowledge extractor, the cross-modal assembly module embeds them into pairs of key and value maps through four dedicated fully connected layers, which serve as projecting the two modal features into the same space and reducing the dimension of features to save computation cost. Specifically, key maps are used for measuring the similarity between visual-modal features and text-modal features. After computing the similarity between the two modal features, we perform a softmax function to generate the final weights for the fine adjustment of the value map of text-modal features, as shown in the following:

$$\widetilde{f}_{\text{text}} = \text{Softmax}\left[\Phi(f_{\text{vis}})^T \Psi(f_{\text{text}})\right] \Theta(f_{\text{text}}) \tag{7}$$

where Φ, Ψ, and Θ denote different fully connected layers. T represents the transpose operation. After adjusting the text-modal features, we concatenate the customized text-modal features $\widetilde{f}_{\text{text}}$ and the value map of RoI visual-modal features, and send them into a fully connected layer to get the fused features, as shown in the following:

$$f_{\text{fused}} = \Omega(\text{Concat}[\Gamma(f_{\text{vis}}), \widetilde{f}_{\text{text}}]) \tag{8}$$

where Ω and Γ denote different fully connected layers. The fused features f_{fused} containing the generalized knowledge of categories are sent to the classification head to predict the category of the object.

The proposed cross-modal assembly module adequately mines the correlation between RoI visual-modal features and class text-modal features through key-value embedding. And it dynamically fuses cross-modal knowledge to enrich the generalized information of categories in original RoI visual features. The generated fused features cover both the RoI visual-modal information and the class text-modal information so that the classification head can deeply use the category-specific knowledge to solve the problem of classification confusion. Compared with the nonlocal self-attention mechanism [69], our proposed module is a cross-attention mechanism applied on cross-modal features. It performs key-value decoupling on both two modal features and completely separates key-value features where key maps are used to measure similarity and value maps store their respective semantic information. And different from DCNet [46] which only fuses visual-modal features of support images and query images, our module fuses RoI visual-modal features and class text-modal features to enhance the generalized and unique characteristics of categories in RoI features.

E. Mask Strategy and Separation Loss

In order that the text knowledge extraction branch can learn to effectively extract comprehensive and useful information from text-modal description sentences, we introduce a mask strategy and a separation loss during training.

1) Mask Strategy: Unlike the long text description paragraphs in zero-shot learning and multiple sentences per image in text-image retrieval, the text-modal descriptions introduced in our proposed model only contain several simple sentences. This means that for each training image, the input of the text knowledge extraction branch is always the same throughout the whole training process, which will inevitably lead to over-fitting. Dropout [70] believes that over-fitting can be alleviated by preventing the co-adaptation of features. Each neuron is invalidated with a certain probability in each forward propagation of training. However, when dropout is combined with the proposed text-modal knowledge extractor, the model can still receive all words in the descriptions and cannot achieve word-level invalidation. In order to guarantee the diversity of text-modal sentences and avoid over-fitting during training, a mask strategy is proposed to perform random word-level masking on text-modal descriptions of each category, as shown in Fig. 6. In each training batch, it randomly masks different parts of words in description sentences of each category, which

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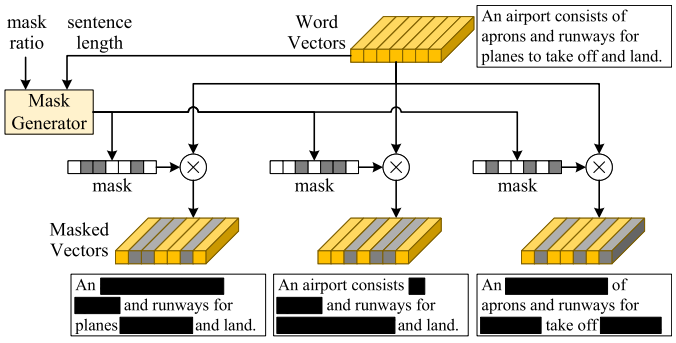


Fig. 6. Illustration of mask strategy. During training, the mask strategy randomly generates mask vectors for each training batch and performs wordwise masks on the description sentences to enhance the diversity of text-modal sentences and avoid over-fitting.

improves the generalization of extracting useful text-modal knowledge from sentences.

According to the mask ratio r and the sentence length T, the mask generator randomly generates mask vectors $\mathcal{M} \in \mathbb{R}^T$ at each training batch. $\mathcal{M}$ is a vector consisting of 0 and 1, where 0 represents the word that needs to be masked and 1 represents the word that is reserved. The number of 0 in $\mathcal{M}$ is [Tr], where $[\cdot]$ means the rounding operation. Then, we multiply the mask vector $\mathcal{M}$ and the word vectors v_t of sentences to generate the masked vectors $\widetilde{v}_t$, as shown in (9). The masked word vectors $\widetilde{v}_t$ are sent to the text-modal knowledge extractor to extract class text features for each category

$$\widetilde{v}_t = \mathcal{M}_t \times v_t, \quad t \in [1, T]. \tag{9}$$

In each training batch, the mask strategy randomly generates different masks for text-modal descriptions, ensuring the diversity of the input text-modal description sentences and preventing over-fitting. During inference, the mask strategy is disabled, and the text-modal knowledge is extracted from the complete description sentences of each category in order to acquire comprehensive characteristics.

2) Separation Loss: In order to fully exploit the role of text-modal sentences in improving generalized information of categories under few-shot scenarios, we expect class text features of different categories to be significantly distinguishable. The class text features should highlight the unique visual-modal features corresponding to each category, rather than the features common to multiple categories. It avoids the confusion of the model caused by highlighting the same features from different categories. In addition, due to the proposed mask strategy, there may be differences in class text features extracted from the descriptions of the same category. We expect that for the sentences of each category, class text features extracted under different masking states can stably highlight the same unique features. Therefore, the extracted class text features of different categories should lead to diverse feature salience, which increases the consistency of the same category and the distinctiveness between different categories.

To this end, we propose a separation loss to diversify the class text features, so that class text features of different categories can stably highlight their own unique and salient visual-modal features. It is well known that different channels

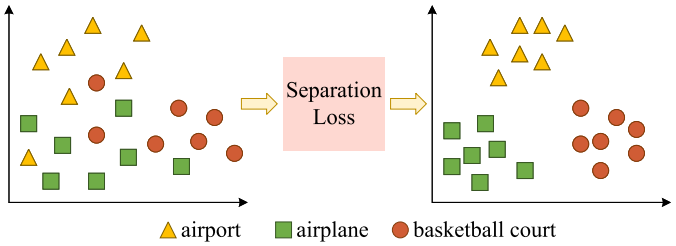


Fig. 7. Illustration of separation loss. Different samples of the same color represent class text features extracted from the description sentences of the same class under different mask vectors. The separation loss constrains the extracted class text-modal features of different categories to be distinguishable and highlights unique visual features.

of RoI visual-modal features are generated by different convolutional kernels and represent different high-level semantic information. The class text features of different categories should be orthogonal and highlight different visual-modal feature channels. Therefore, the proposed separation loss is implemented by an orthogonal loss as shown in (10). It constrains the orthogonality between class text features of different categories, which ensures that the class text features of different categories are separable and the class text features of the same category are consistent in terms of highlighting visual-modal features

$$\mathcal{L}_{\text{sepa}} = \frac{1}{C} \sum_{i=1}^{C} \sum_{j \neq i} \left| f_{\text{text},i}^{T} f_{\text{text},j} \right| \tag{10}$$

where C is the number of classes, and $|\cdot|$ means the absolute value. $\mathcal{L}_{\text{sepa}}$ is nonnegative. The larger the $\mathcal{L}_{\text{sepa}}$, the more similar the f_{text} between different classes, and the poorer distinctiveness of f_{text}. On the contrary, the smaller the $\mathcal{L}_{\text{sepa}}$, the more orthogonal the f_{text} between different classes, and the stronger the distinctiveness of f_{text}. When $\mathcal{L}_{\text{sepa}}$ is equal to 0, it means that f_{text} of all categories are orthogonal to each other, and the distinctiveness of f_{text} is the strongest.

As shown in Fig. 7, for the class text features extracted from different masked description sentences, separation loss makes the features of the same class more aggregated and the features belonging to different classes easier to distinguish. With the constraint of separation loss, the ambiguity between the text-modal knowledge of different categories is greatly reduced, and the class text feature of each category has its own particularity.

IV. Experiments

In this section, we first briefly introduce three datasets for experiments. Then, the evaluation metrics and implementation details are explained. Finally, we analyze the experimental results of our proposed TEMO and compared methods.

A. Datasets

1) DIOR [2]: DIOR is a large-scale object detection benchmark of RSIs. It contains 23 463 images collected from Google Earth with 192 472 object instances. The objects are labeled into 20 categories, including airplane, airport, baseball field, basketball court, bridge, chimney, dam, expressway service

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area, expressway toll station, harbor, golf course, ground track field, overpass, ship, stadium, storage tank, tennis court, train station, vehicle, and wind mill. The size of the images is 800 pixels, and the spatial resolutions range from 0.5 to 30 m.

2) NWPU VHR-10 [13]: NWPU VHR-10 consists of 715 high-spatial-resolution RGB images from Google Earth and 85 pan-sharpened color infrared images from the Vaihingen dataset [71]. It contains 650 positive images with 3775 object instances and 150 negative images without objects. The dataset has ten geospatial object classes, including airplane, baseball diamond, basketball court, bridge, harbor, ground track field, ship, storage tank, tennis court, and vehicle.

3) FAIR1M [3]: FAIR1M is a benchmark for fine-grained object detection. It contains more than 15 000 images with over 1.02 million objects from Gaofen satellites and the Google Earth platform. Objects are annotated into five categories and 37 subcategories. The size of images varies from 1000 pixels to 10 000 pixels, with spatial resolutions ranging from 0.3 to 0.8 m. To verify the effectiveness of our method in fine-grained object detection, we select the airplane category with explicit types (FAIR1M-Airplane) for experiments, which contains 11 subcategories with Boeing 737, Boeing 747, Boeing 777, Boeing 787, Airbus A220, Airbus A321, Airbus A330, Airbus A350, COMAC C919, COMAC ARJ21, and other airplane. Additionally, since our research aims to improve the performance of FSOD, rather than rotated detection, we convert rotation annotations into horizontal annotations.

B. Evaluation Metrics

The same as few-shot object detection model (FSODM) [12], the mean average precision (mAP) of novel classes is applied as the evaluation metric to verify the performance of FSOD methods. First, based on the intersection of union (IoU) of the predicted boxes and ground truth, the number of samples with true positive (TP) for correctly predicting, false positive (FP) for wrongly matching the background as foreground, and false negative (FN) for undetected objects are calculated. Then, the precision and recall are calculated as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{11}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{12}$$

Finally, according to the precision (P) and recall (R) of each category, the P-R curve can be plotted, and the area of the P-R curve is calculated as the AP for each category

$$\text{AP} = \int_0^1 \text{PdR}. \tag{13}$$

Following PASCAL VOC [72], mAP is obtained as the average of AP of each class to evaluate the performance of all novel classes comprehensively.

C. Implementation Details

In order to verify the performance of object detection in few-shot scenarios, we divide the classes of each dataset into base classes and novel classes. The novel classes of datasets are shown in Table I, and the other classes are used as base classes. For DIOR and NWPU VHR-10, we adopt the same division as FSODM [12], selecting five classes and three classes as novel classes, respectively. For FAIR1M-Airplane, we select three classes (A350, Boeing737, and Boeing747) as novel classes and the other eight classes as base classes. In the experiments, we simulate k-shot scenarios, i.e., each novel class has k object instances. And k is set to 2, 3, 5, and 10.

TABLE I
NOVEL CLASSES IN EACH DATASET FOR FEW-SHOT SCENARIOS

Data Set	Novel Classes
DIOR [2]	airplane, baseball field, tennis court, train station, windmill
NWPU VHR-10 [13]	airplane, baseball diamond, tennis court
FAIR1M-Airplane [3]	A350, Boeing737, Boeing747

For a fair comparison with other methods, we apply the same multiscale training strategy as FSODM [12]. And random horizontal flipping and normalization are used for training. For DIOR and NWPU VHR-10, the original images are used directly. For FAIR1M-Airplane, we crop the original image to 800 × 800 pixels with 200-pixel overlap and finally acquire 12 964 training patches and 5413 test patches. To demonstrate the effectiveness of our method, we compare our method with some state-of-the-art FSOD methods proposed for natural scene images [5], [6], [7], [10], [11], [45] and specifically designed for RSIs [12], [49]. In addition, to reduce the effect of different support images and ensure the credibility of the results, the record results of our method and some reproduced methods are the average of 20 random experiments. In each experiment, different images are randomly selected for each novel class to train the model in the few-shot fine-tuning stage.

Our TEMO uses faster RCNN [14] with ResNet-101 [73] pretrained on ImageNet [74] as the base detector. According to the value of losses, the balance factor λ in (1) is set to 0.01 to make the separation loss and the detection loss comparable in magnitude. The model is optimized by the stochastic gradient descent (SGD) optimizer with a momentum of 0.9 and a weight decay of 0.0001. We set the batch size as 2 on a 16-GB NVIDIA Tesla P100 GPU. For base model training, the model is trained for 85k, 15k, and 10k iterations with learning rates of 0.005, 0.0005, and 0.00005. And for the few-shot fine-tuning stage, the model is trained for 5000, 2000, and 1000 iterations with the learning rates of 0.005, 0.0005, and 0.00005, respectively.

D. Ablation Studies

To demonstrate the effectiveness of our proposed TEMO and its components, we conduct sufficient ablation experiments on the DIOR dataset.

1) Ablation Study for Strong Baseline: To verify the performance of our proposed baseline, we freeze different layers of the detector during the second few-shot fine-tuning stage. The experimental results are presented in Table II. TFA with excellent performance on natural scene images, freezing all layers

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TABLE II
ABLATION STUDY FOR THE PROPOSED STRONG BASELINE

Frozen Layers	Shot			
	2	3	5	10
B+F+R+RA (TFA)	7.0	8.5	10.6	11.8
B+F+R	14.5	15.0	17.9	21.3
B+F	22.9	23.1	23.9	31.8
B (our baseline)	**23.4**	**25.9**	**30.9**	**34.2**
None	19.3	17.2	17.3	26.1

* Abbreviation of frozen layers - B: Backbone; F: FPN; R: RPN; RA: RoI Align.

TABLE III
ABLATION STUDY FOR DIFFERENT COMPONENTS

Components				Shot			
TMK	CMA	MS	SL	2	3	5	10
				23.4	25.9	30.9	34.2
✓	✓			26.5	29.3	34.1	38.6
✓	✓	✓		27.1	30.8	36.7	41.9
✓	✓		✓	26.9	30.2	36.8	40.7
✓	✓	✓	✓	**27.5**	**30.9**	**37.2**	**42.8**

* Abbreviation of components - TMK: Text-Modal Knowledge Extractor; CMA: Cross-Modal Assembly Module; MS: Mask Strategy; SL: Separation Loss.

except the last layer of the prediction head, does not perform well on RSIs. When the RoI align layer and RPN are gradually unfrozen and fine-tuned with the prediction head together, the performance of the model on novel classes is significantly improved. It proves that for novel classes, fine-tuning the RoI align layer and RPN can extract high-quality proposals and representative features. Furthermore, our proposed baseline improves the performance under all shot settings by only freezing the backbone and fine-tuning all other layers, which illustrates the necessity of fine-tuning FPN. However, when fine-tuning the entire detector including the backbone, the over-fitting problem occurs due to the lack of training data, and the performance of novel classes drops considerably.

These experimental results illustrate the effectiveness of our proposed baseline for FSOD in RSIs. Furthermore, we find that the errors of the proposed baseline for novel classes are mainly in the classification branch. It is impossible to extract generalized and unique features for categories from limited data. Hence, we introduce text-modal knowledge that highly generalizes the characteristics of each category to strengthen the information of categories in visual-modal features.

2) Ablation Study for Each Component: Table III shows the effectiveness of the proposed components in TEMO. By introducing text-modal knowledge, that is, adding text-modal knowledge extractor and cross-modal assembly module, the detection performance for novel classes is significantly improved, with more than 3% improvement of mAP in all shot settings. Additionally, the mask strategy and the separation loss are added, respectively, to avoid over-fitting and ambiguity of the extracted text-modal features, and the performance is further improved. When applying both mask strategy and separation loss, the model achieves the highest performance. Compared with the proposed baseline, TEMO improves by 4.1%, 5.0%, 6.3%, and 8.6% in 2-, 3-, 5-, and 10-shots, respectively.

TABLE IV
ABLATION STUDY FOR TEXT-MODAL KNOWLEDGE EXTRACTOR

Extracted Block	Shot			
	2	3	5	10
RNN [75]	25.5	27.8	33.0	37.2
LSTM [76]	26.1	28.8	33.5	38.5
GRU [68]	**26.5**	**29.3**	34.1	38.6
Transformer [77]	26.0	29.1	**34.7**	**38.9**

TABLE V
ABLATION STUDY FOR WORD EMBEDDING LAYERS

Word Embedding Layers	Shot			
	2	3	5	10
word2vec [78]	26.8	29.7	37.0	41.9
GloVe [79]	27.0	30.1	36.5	42.6
skip-thought [67]	27.5	**30.9**	**37.2**	42.8
CLIP [80]	**27.7**	30.4	36.8	**43.2**

The experimental results show that through the text-modal knowledge extractor and cross-modal assembly module, introducing text-modal knowledge into visual-modal features greatly improves the performance of novel classes. Further use of the introduced mask strategy and separation loss can enhance the diversity of text-modal knowledge and reduce the ambiguity of descriptions between different categories.

3) Ablation Study for Text-Modal Knowledge Extractor: To verify the robustness of the text-modal knowledge extractor, we use different text processing blocks to extract text-modal knowledge from text descriptions. The experimental results are shown in Table IV. The performance of different processing blocks is similar, illustrating the robustness of the text-modal knowledge extractor to different processing blocks. Among them, due to the weak long-term memory of recurrent neural network (RNN [75]), the knowledge extracted from sentences is not comprehensive, so the performance is slightly lower. Using long short-term memory (LSTM [76]) and GRU [68] improves the performance because they enhance the long-term memory and utilize the information of the entire sentences. Furthermore, transformer block [77] exploits global self-attention to capture long dependencies of entire descriptions and slightly improves the performance. Taking into account both the performance and computation cost, GRU is selected as the processing block for other experiments.

Additionally, to verify the robustness of our method on different word embedding layers, we use four common pretrained word embedding layers and the corresponding encoders to generate class text features. The experimental results are shown in Table V. Since word embedding layers are loaded as initialization and are updated according to the input sentences and images during training, there is a minor difference in the performance of different word embedding layers when using the same encoder structure. And for CLIP, using a transformer as the encoder, the performance difference is still subtle. It demonstrates that the proposed method is robust to different word embedding layers.

4) Ablation Study for Cross-Modal Assembly Module: We apply different assembly methods to fuse the extracted class text-modal features and RoI visual-modal features, as shown

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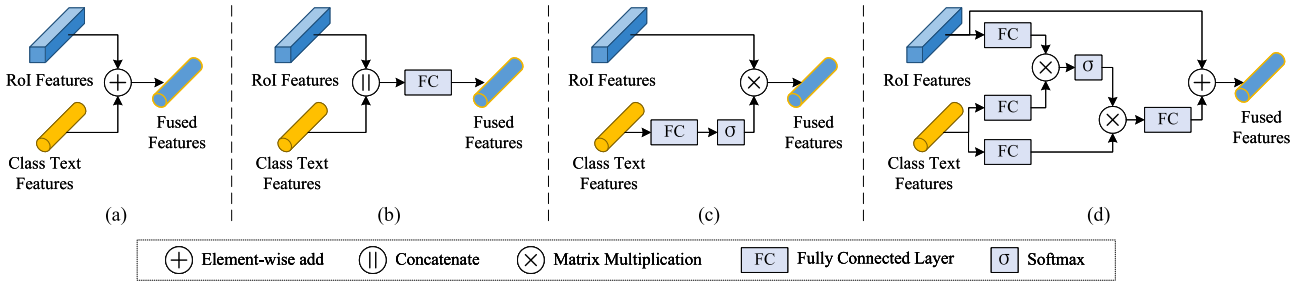


Fig. 8. Different assembly methods for fusing RoI visual-modal features and class text-modal features to generate fused features. (a) Add. (b) Concat. (c) Vanilla attention. (d) Nonlocal.

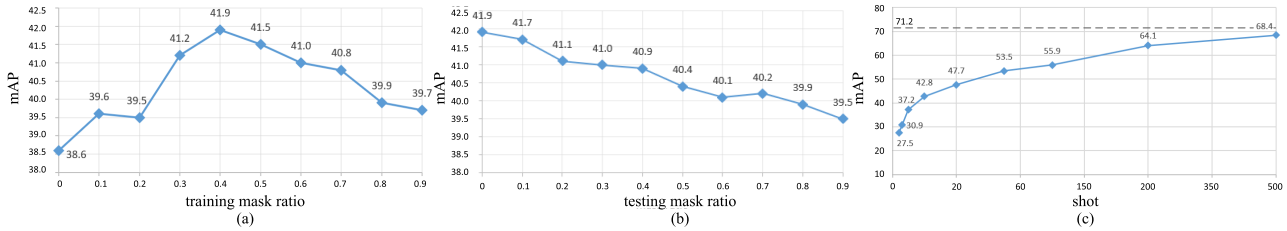


Fig. 9. Ablation study for (a) training mask ratio, (b) testing mask ratio, and (c) number of shots. The blue line represents the results of our FSOD method, and the black dashed line represents the upper bound of the performance on novel classes trained with all data.

TABLE VI
ABLATION STUDY FOR CROSS-MODAL ASSEMBLY MODULE

Assembly Method	Shot			
	2	3	5	10
(a) Add	24.7	28.1	32.2	37.0
(b) Concat	24.9	28.5	32.8	37.4
(c) Vanilla Attention	25.2	28.6	33.5	37.9
(d) Non-local	25.3	28.8	33.6	38.0
Ours	**26.5**	**29.3**	**34.1**	**38.6**

TABLE VII
ABLATION STUDY FOR THE MEASURES TO AVOID OVER-FITTING

Ways to Avoid Over-fitting	Shot			
	2	3	5	10
None	26.5	29.3	34.1	38.6
Dropout [70] ($p = 0.5$)	26.8	30.2	34.9	40.2
Mask Strategy	**27.1**	**30.8**	**36.7**	**41.9**

TABLE VIII
ABLATION STUDY FOR DIFFERENT DESCRIPTIONS

Descriptions	Shot			
	2	3	5	10
1	27.5	30.9	37.2	42.8
2	27.4	31.2	36.8	42.8
3	27.1	30.6	37.1	42.5

in Fig. 8. The experimental results are presented in Table VI. It can be seen that simply adding or concatenating text-modal features and visual-modal features may cause a puzzle between the two modal information, resulting in insignificant performance improvement. Through the attention mechanism, the vanilla attention and nonlocal use text-modal information to enhance the corresponding RoI visual-modal features, thereby improving the discriminative information of visual features and reducing the classification confusion. Although the performance is improved, these two attention mechanisms directly fuse the two modal features, and the fused features are prone to fusion deviation and contain numerous noises, resulting in the submersion of prior information. Furthermore, our proposed cross-modal assembly module achieves the highest performance on all shot settings. It decouples the similarity measure and the retention of the useful information by completely separating the key-value pairs of the two modal features. Thus, the text-modal information can be applied to accurately emphasize the relevant information representation in the visual-modal features, which preserves the discriminative information of both text-modal features and visual-modal features.

5) Ablation Study for Mask Strategy: To verify the effectiveness of the proposed mask strategy for word-level masking, we compare dropout with the mask strategy. The results are shown in Table VII. Compared with no measure to avoid over-fitting, using dropout improves the average performance of all settings by 0.9%. And compared with dropout, using the mask strategy further improves the average performance by 1.1%. Experimental results demonstrate the effectiveness and justification of the proposed mask strategy.

Besides, we conduct experiments with different training mask ratios in a 10-shot setting on the DIOR dataset to determine an appropriate mask ratio for text-modal descriptions during training. The results are shown in Fig. 9(a). It is found that as long as the mask strategy is applied, no matter what the mask ratio is set to, the detection performance of novel classes is greatly improved than that without the mask strategy (mask ratio = 0). As the mask ratio increases, the diversity of the extracted text-modal knowledge gradually increases, which effectively prevents the over-fitting of the text knowledge extraction branch. The performance achieves optimal when the mask ratio is 0.4. When the mask ratio exceeds 0.4, since too many words are masked out and the information is insufficient, the performance decreases slowly with the increase in mask ratio. But even if masking off 90% of the text descriptions, the model still achieves favorable performance.

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TABLE IX
FSOD RESULTS ON DIOR DATASET

Shot	Methods	Novel Classes					mAP
		airplane	baseball field	tennis court	train station	windmill	
2	Meta RCNN† [7]	9.1	15.4	33.6	10.1	4.3	14.5
	FsDetView† [45]	9.1	9.1	18.0	9.1	0.1	9.1
	TFA† [10]	12.0	24.6	33.7	8.9	12.2	18.3
	FSCE† [11]	20.7	38.5	42.2	10.0	14.0	25.1
	TEMO†(ours)	**24.4**	**43.0**	**43.1**	**11.4**	**15.4**	**27.5**
3	Meta RCNN† [7]	9.1	17.8	33.5	3.8	9.1	14.7
	FsDetView† [45]	9.1	17.9	18.1	9.1	9.1	12.6
	TFA† [10]	11.9	22.6	51.8	9.7	8.1	20.8
	PAMS-Det [49]	14.0	**54.0**	24.0	17.0	**31.0**	28.0
	FSCE† [11]	17.3	38.7	40.8	11.6	16.0	24.9
	TEMO†(ours)	**18.4**	47.1	**52.4**	**17.7**	18.8	**30.9**
5	Meta RCNN† [7]	9.1	25.9	32.7	7.0	2.3	15.4
	FsDetView† [45]	9.1	22.3	26.3	5.0	1.1	12.8
	TFA† [10]	13.2	27.7	51.8	9.0	8.0	21.9
	Meta YOLO [5]	9.0	33.0	47.0	9.0	13.0	22.0
	RepMet [6]	9.0	19.0	11.0	1.0	1.0	8.0
	FSODM [12]	9.0	27.0	**57.0**	11.0	19.0	25.0
	PAMS-Det [49]	17.0	55.0	41.0	17.0	**34.0**	33.0
	FSCE† [11]	22.8	36.6	49.5	17.1	15.2	28.3
	TEMO†(ours)	**24.2**	**58.9**	52.1	**25.0**	25.7	**37.2**
10	Meta RCNN† [7]	12.4	37.5	32.4	14.0	3.2	19.9
	FsDetView† [45]	9.1	31.7	25.8	11.1	6.1	16.7
	TFA† [10]	17.7	33.1	56.4	17.3	8.7	26.7
	Meta YOLO [5]	15.0	45.0	54.0	7.0	18.0	28.0
	RepMet [6]	13.0	33.0	24.0	1.0	1.0	14.0
	FSODM [12]	16.0	46.0	60.0	14.0	24.0	32.0
	PAMS-Det [49]	25.0	58.0	50.0	23.0	**36.0**	38.0
	FSCE† [11]	24.7	56.2	49.1	21.6	21.4	34.6
	TEMO†(ours)	**32.6**	**59.3**	**67.0**	**23.9**	31.2	**42.8**

† represents the average results over 20 random experiments of the methods we reproduced.

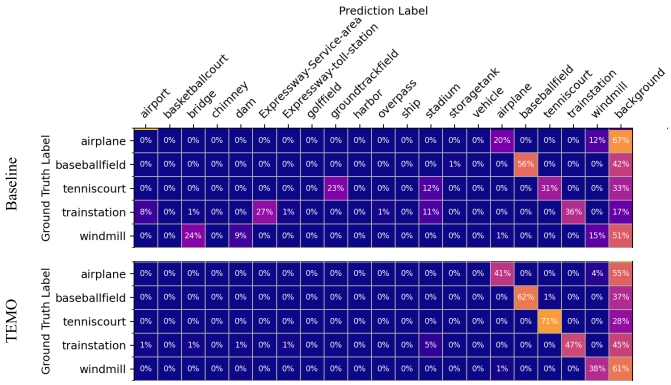


Fig. 10. Classification confusion matrix of our proposed baseline (top) and TEMO (bottom) on DIOR dataset with 10-shot setting.

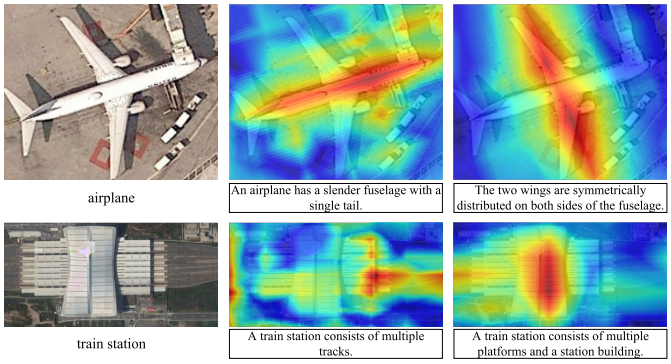


Fig. 11. Enhanced RoI visual features with different text-modal descriptions on DIOR dataset with 10-shot setting.

It proves that the text-modal descriptions can highly summarize the generalized and unique characteristics of categories, and introducing text-modal knowledge can effectively enhance the insufficient generalized information of each category in visual-modal features and improve the performance of novel classes. We choose the mask ratio of 0.4 in other experiments.

6) Ablation Study for Different Sentences: To verify the impact of different descriptions on the performance of our model, we conduct ablation experiments on testing mask ratio and different sentences. First, we choose the mask ratio with the best performance as 0.4 during training and then use different mask ratios for testing. The results are shown in Fig. 9(b). It shows that introducing the whole description during testing achieves the best performance. As the testing mask ratio increases, the information brought by text-modal descriptions gradually decreases, so the model performance gradually decreases. Compared with using all descriptions, the performance drops by only 2.6% when masking out 90% of text-modal descriptions, which is still ahead of the baseline. Additionally, we apply different sentences of novel classes to verify the robustness of our model under different descriptions. For the five novel classes in the DIOR dataset, we invite different experts to summarize three kinds of sentences and

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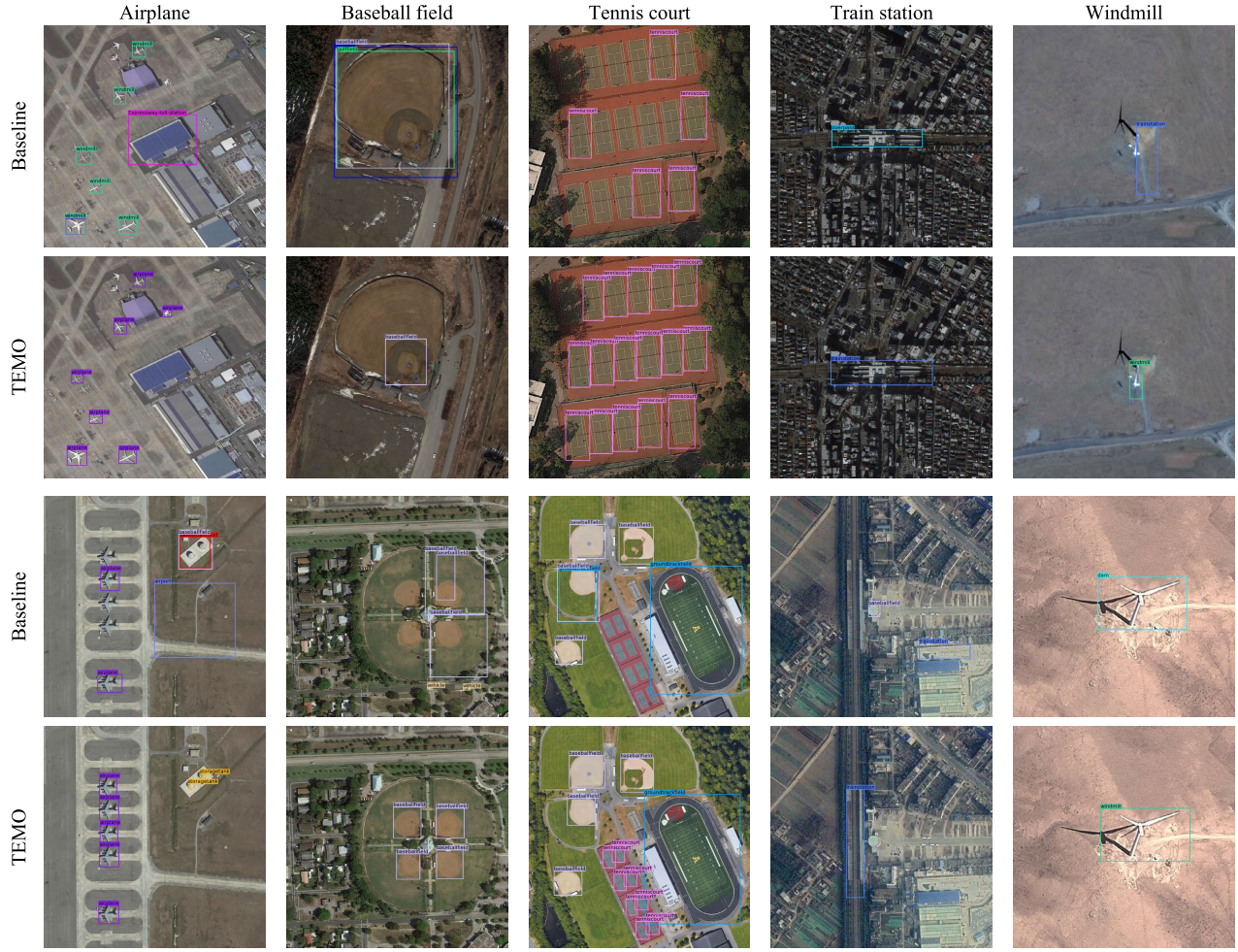


Fig. 12. Visualization comparison between our proposed baseline and TEMO on DIOR dataset of 10-shot setting. Only the bounding boxes with a confidence greater than 0.3 are visualized.

apply these different descriptions to fine-tune the model well-trained in the first stage. The results are shown in Table VIII. For different sentences, the changes in the performance are not obvious, indicating the robustness and stability of the proposed method to different description sentences. Thus, our proposed model is robust to the text-modal descriptions used in training.

7) Ablation Study for the Number of Shots: As we gradually increase the number of shots, the trend of our model's performance is shown in Fig. 9(c). The black dashed line is the result of the novel classes of the model trained with the complete data, where the number of instances of the five novel classes is 1892, 2384, 4906, 502, and 2365, respectively. It achieves 71.2% mAP on novel classes, representing the upper bound of the performance for FSOD methods. We conduct experiments with k-shots ($k = 2, 3, 5, 10, 20, 50, 100, 200, 500$), respectively. As the number of shots increases, the performance of TEMO gradually improves. When the number of shots is small, the performance improves rapidly as the shot increases. The mAP of novel classes increases from 27.5% to 47.7% when the number of shots increases from 2 to 20. When the number of shots is increased to 50 and 100, the mAP reaches 53.5% and 55.9%, respectively. And the performance achieves 64.1% when the shot further increases to 200. When the number of shots continues to increase to 500, the performance reaches 68.4%, only a drop of 2.8% when the number of training samples is about one-fifth of the total data. It shows that introducing text-modal knowledge provides generalized information of categories, and the satisfactory performance is achieved with a few training samples.

8) Ablation Study for Qualitative Analysis: In order to intuitively show the impact of introducing text-modal knowledge for FSOD, we present the classification confusion matrix of the novel classes under a 10-shot setting between the proposed baseline and TEMO in Fig. 10. Through introducing text-modal knowledge, the classification accuracy of our TEMO for novel classes is greatly improved, which effectively reduces the error of misclassifying novel classes as base classes. It demonstrates that the introduced text-modal knowledge strengthens the generalized and unique characteristics of each category in visual-modal features, and alleviates the classification confusion to improve the performance of FSOD. It also can be seen that the main mistake of our model is to classify the novel classes as background. Due to the complex background in RSIs, the text-modal knowledge cannot summarize the characteristics of the background in a few descriptive sentences. Some object instances will be misclassified as the background when facing complex background, resulting in missed detection. This is an issue to be studied in our future

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TABLE X
FSOD RESULTS ON NWPU VHR-10 DATASET

Methods	Shot															
	2				3				5				10			
	PL	BD	TC	mAP	PL	BD	TC	mAP	PL	BD	TC	mAP	PL	BD	TC	mAP
Meta RCNN† [7]	0.6	11.9	6.1	6.2	6.8	7.1	5.7	6.5	12.0	21.9	0.9	11.6	11.4	24.4	10.5	15.4
FsDetView† [45]	9.1	21.6	4.5	11.7	9.1	22.9	2.4	11.5	16.5	28.4	7.8	17.6	9.1	20.0	12.4	13.8
TFA† [10]	9.1	12.4	14.9	12.1	9.1	16.2	14.5	13.2	9.1	15.9	24.6	16.5	4.5	17.0	21.9	14.5
Meta YOLO [5]	-	-	-	-	13.0	12.0	11.0	12.0	24.0	39.0	11.0	24.0	20.0	74.0	26.0	40.0
RepMet [6]	-	-	-	-	19.0	36.0	12.0	23.0	20.0	36.0	14.0	23.0	22.0	39.0	18.0	26.0
FSODM [12]	-	-	-	-	15.0	57.0	25.0	32.0	58.0	84.0	16.0	53.0	60.0	88.0	48.0	65.0
PAMS-Det [49]	-	-	-	-	21.0	76.0	16.0	37.0	55.0	**88.0**	20.0	55.0	61.0	88.0	50.0	66.0
FSCE† [11]	17.6	51.0	20.5	29.7	33.3	73.4	30.2	45.6	60.7	83.6	32.5	59.0	85.2	75.3	48.0	69.5
TEMO†(ours)	**54.4**	**56.6**	**21.5**	**44.2**	**49.1**	**77.8**	**45.8**	**57.6**	**71.4**	85.4	**43.5**	**66.8**	**86.4**	**88.7**	**50.1**	**75.1**

* Abbreviation of novel classes - PL: airplane; BD: baseball diamond; TC: tennis court. † represents the average results over 20 random experiments of the methods we reproduced.

work. Furthermore, we show the enhanced RoI visual-modal features with different text-modal descriptions in Fig. 11. Our TEMO can enhance the corresponding parts of RoI features according to the text-modal descriptions instead of simply memorizing the characteristics of categories, which indicates that our model has learned to extract knowledge from the text-modal descriptions to enhance visual-modal features.

E. Comparisons With Sate-of-the-Art Methods

1) Experiments on DIOR: We compare our model with state-of-the-art methods on the DIOR dataset, including 2-, 3-, 5-, and 10-shots. The experimental results are shown in Table IX. It can be seen that with the increase in the number of training instances k of novel classes, the performance of almost all FSOD methods gradually improves. And as we stated in Section I, meta-learning methods in natural scenes [6], [7], [45], which spend lots of effort in learning the transferability between episodes, perform lower than fine-tuning-based methods [10], [11]. FSODM [12] imparts a multiscale strategy to the meta-learning method to improve the performance on RSIs, but it is still lower than the fine-tuning-based method PAMS-Det [49] which is also designed for RSIs. Compared to these FSOD methods, TEMO achieves the highest performance, with mAPs for novel classes reaching 27.5%, 30.9%, 37.2%, and 42.8% in 2, 3, 5, and 10-shot scenarios, respectively. The results prove the effectiveness of our proposed method of FSOD in remote sensing.

We visually demonstrate the results of the proposed baseline and TEMO on novel classes in Fig. 12. It can be seen that the baseline can locate the objects belonging to novel classes. Still, due to the insufficient generalized information for categories in visual-modal features, it misclassifies these novel objects as the base classes or background, resulting in misclassification or missed detection. For example, the wings of some airplanes are seen as the blades of windmills, and some airplanes with unusual appearances are recognized as backgrounds. However, our method strengthens the information of categories in visual features by introducing text-modal knowledge, which alleviates classification confusion effectively.

2) Experiments on NWPU VHR-10: Table X presents the results of our TEMO and other compared methods on the NWPU VHR-10 dataset. It can be seen that the performance of the natural scene FSOD methods on the NWPU VHR-10 dataset is similar to the DIOR dataset, i.e., the performance of meta-learning-based methods is lower than that of fine-tuning-based methods. By applying a multiscale strategy, FSODM greatly improves the performance of RSIs. PAMS-Det embellishes the fine-tuning-based method for RSIs and further improves the performance of novel classes. Compared with these methods, our method achieves substantial performance gains under all shot settings. And as the number of shots increases, the performance of our model improves rapidly. For example, from 2-shot to 3-shot, the mAP is improved by 13.4%.

We visualize the detection results of the proposed baseline and TEMO for the three novel classes under the 10-shot setting, as shown in Fig. 13. The baseline suffers from many

TABLE XI
FSOD RESULTS ON FAIR1M-AIRPLANE DATASET

Shot	Methods	Novel Classes			mAP
		A350	B737	B747	
2	Meta RCNN† [7]	18.5	9.1	46.1	24.6
	FsDetView† [45]	20.2	9.1	46.6	25.3
	TFA† [10]	20.3	15.6	**50.8**	28.9
	FSCE† [11]	21.4	16.2	43.1	26.9
	TEMO†(ours)	**26.2**	**27.5**	38.1	**30.6**
3	Meta RCNN† [7]	19.2	15.5	**48.7**	27.8
	FsDetView† [45]	24.3	17.5	47.3	29.7
	TFA† [10]	25.1	**29.1**	39.2	31.1
	FSCE† [11]	21.9	25.3	38.2	28.6
	TEMO†(ours)	**26.0**	27.3	45.1	**32.8**
5	Meta RCNN† [7]	23.9	12.7	49.8	28.8
	FsDetView† [45]	25.9	12.8	50.7	29.8
	TFA† [10]	25.6	28.6	37.8	30.7
	FSCE† [11]	22.3	18.5	**59.7**	33.5
	TEMO†(ours)	**26.0**	**33.2**	52.9	**37.4**
10	Meta RCNN† [7]	22.4	32.7	52.1	35.7
	FsDetView† [45]	23.9	24.9	55.0	34.6
	TFA† [10]	23.1	33.2	34.8	30.4
	FSCE† [11]	24.0	16.8	61.1	34.0
	TEMO†(ours)	**24.7**	**36.5**	**61.3**	**40.8**

* Abbreviation of novel classes - A350: A350; B737: Boeing737; B747: Boeing747. † represents the average results over 20 random experiments of the methods we reproduced.

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Fig. 13. Visualization comparison between our proposed baseline and TEMO on NWPU VHR-10 dataset of 10-shot setting. The red ellipses identify the improved areas of our TEMO compared with the baseline. Only the bounding boxes with a confidence greater than 0.3 are visualized.

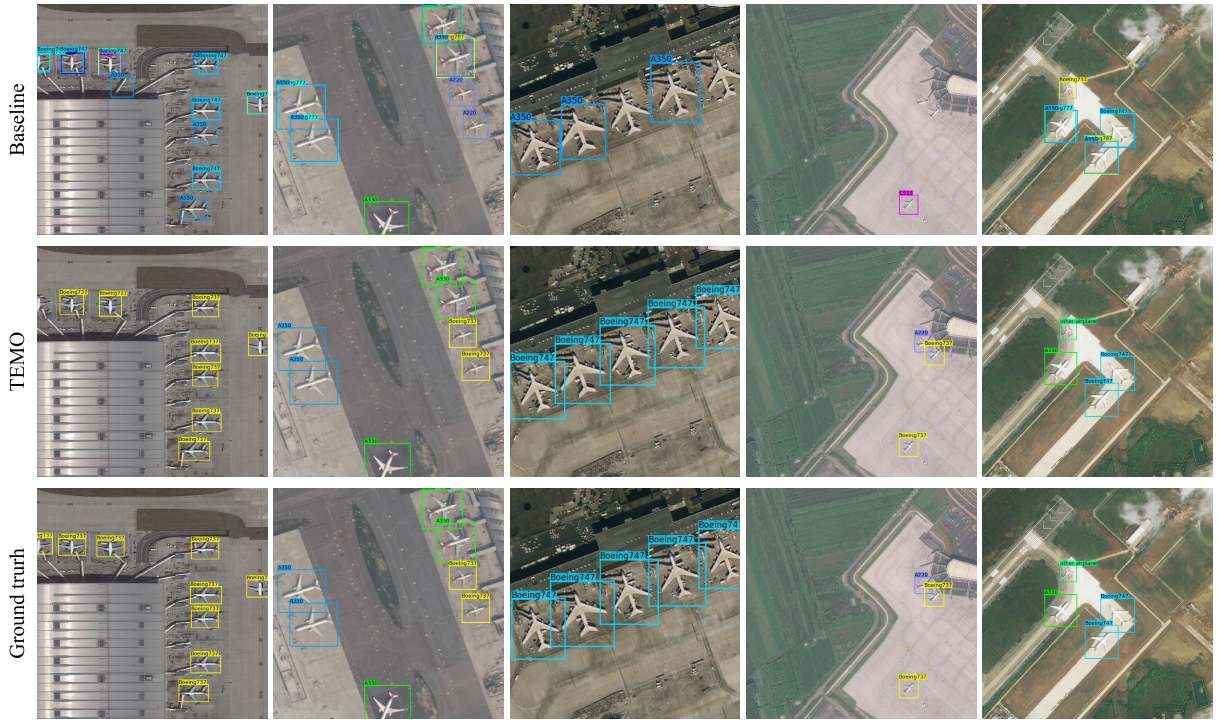


Fig. 14. Visualization comparison between our proposed baseline and TEMO on FAIR1M-Airplane dataset of 10-shot setting. Only the bounding boxes with a confidence greater than 0.3 are visualized.

misclassifications, i.e., misclassifying novel classes as base classes or backgrounds. For example, the baseline misclassifies airplanes and tennis courts as basketball courts. However, our method is able to extract the generalized and unique features of categories and improves the performance of novel classes.

3) Experiments on FAIR1M-Airplane: We also conduct experiments on the FAIR1M-Airplane dataset to validate the performance of our model for fine-grained object detection under few-shot conditions. The experimental results of our method and some reproduced methods are shown in Table XI. Unlike the above-mentioned two object detection datasets, meta-learning-based methods perform on par with fine-tuning-based methods on the FAIR1M-Airplane dataset and even outperform fine-tuning-based methods in some shot settings. Due to the fact that the differences between different episodes are subtle for fine-grained object detection, the meta-learning-based methods do not need to spend too much effort learning the transferability of different episodes. They concentrate on learning the representation of categories to improve the performance. Nonetheless, our method improves the performance of fine-tuning-based methods by introducing text-modal knowledge, which still achieves the highest performance in novel classes. We also visualize the results of our model in Fig. 14. It can be seen that TEMO effectively identifies the type of novel aircraft. Experimental results on FAIR1M-Airplane demonstrate that our model is also effective for fine-grained object detection.

V. Conclusion

In this article, we first analyze why fine-tuning-based FSOD methods outperform meta-learning-based methods in RSIs and

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TABLE XII
TEXT-MODAL DESCRIPTION SENTENCES OF EACH CATEGORY IN DATASETS

Data Sets		Classes	Text-modal Descriptions
DIOR	NWPU	airplane	An airplane includes fuselage, wings, tail, and engines, etc. The two wings of the aircraft are symmetrically distributed on both sides of the fuselage.
		baseball diamond	A baseball diamond is a quarter circle and usually comes in pairs. It is covered by grass.
		basketball court	The basketball court is usually close to ground track fields and consists of a rectangular floor with baskets at each end. Outdoor surfaces are generally made from standard paving materials.
		bridge	A bridge is usually slender. It is a structure built to span a physical obstacle such as a body of water, valley, road, or rail, without blocking the way underneath.
		ground track field	A ground track field contains the outer oval shaped running track and an area of turf within this track.
		harbor	A harbor is a sheltered body of water. It provides docking areas for ships, boats, and barges.
		ship	Ships are usually found in oceans, rivers, lakes, etc. where there is water. The shape of a ship is a long bar with pointed ends.
		storage tank	Storage tanks are usually vertical cylindrical. They are white or gray. Multiple tanks usually appear together.
		tennis court	A tennis court is the venue where the sport of tennis is played. It is a firm rectangular surface with a low net stretched across the centre.
		vehicle	The vehicle contains four tires and has windshields front and rear. Vehicles drive on the road or are stationed in the parking lot.
		airport	An airport consists of a landing area with airport aprons and runways for planes to take off and to land. It includes adjacent utility buildings such as control towers, hangars and terminals.
		chimney	Chimneys made of masonry, clay or metal are usually conical in shape, large below and small above. They are typically vertical, or as close to vertical as possible.
		dam	Dams are usually built along the water edge, narrow at the top and wide at the bottom. They are barriers that prevent or restrict the flow of surface water or underground streams.
		Expressway-Service-area	An expressway service area is usually distributed on both sides of the road. It includes parking lots and buildings for people to rest and shopping.
		Expressway-toll-station	An expressway toll station is an enclosure placed along a toll road and consists of several adjacent toll booths and spans the entire highway.
		golf field	A golf field is halfway up the mountain and usually covers a wide area with many fairways intertwined.
		overpass	An overpass is a bridge or road that crosses over another road or railway. An overpass and underpass together form a grade separation.
		stadium	Stadiums are usually open air sports venues. It is round or oval. It contains auditoriums, a running track, and a football pitch.
		train station	A train station generally consists of multiple platforms, multiple tracks and a station building. Trains stop at the train station.
		windmill	A windmill is a structure that contains three blades which are erected by a tall column.
FAIR1M-Airplane		A220	The A220 is a narrow body aircraft. The fuselage is medium in size, the wings are curved, the overall color is the same, and the tail is serrated.
		A321	The A321 is a narrow body aircraft with a medium to large fuselage of special size. The wings are swept back by 25 degrees. It is powered by two wing pylon-mounted turbofan engines with a single vertical stabilizer and rudder.
		A330	The A330 has a large fuselage, the wings are curved. There are two black lines at the position of the wings corresponding to the engine, and two black patches near the outer side.
		A350	The A350 has a large fuselage with curved wings. There are two black patches on the outside of the engines and two vertical protrusions on the inside of the engines.
		ARJ21	The engine of the ARJ21 is located at the rear of the fuselage. The wings are straight, and the outermost side has an inner hook. There are bumps on both sides of the rear of the fuselage.
		Boeing737	Boeing737 is a twin engine narrow body aircraft on short and thin routes. It has straight wings. The fuselage is dark, and the edges and sides near the fuselage are white.
		Boeing747	Boeing 747 is a large, long range wide body airliner with a large fuselage of swept wings and four engines.
		Boeing777	Boeing 777 adopts a round fuselage design with straight wings, dark overall color, white edges, and no serrations on the inside.
		Boeing787	Boeing787 is a wide body jet airliner. The fuselage is 57-68 meters, and the wingspan is 60 meters. The wings have straight lines, darker overall color, white edges, and serrations on the inside.
		C919	The wings of the C919 are curved and the tail is serrated. The color of the tail is relatively uniform, generally blue and green.
		other-airplane	An airplane does not fall into the listed categories and includes fuselage, wings, tail, and engines, etc. The two wings of the aircraft are symmetrically distributed on both sides of the fuselage.

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propose a strong baseline based on the advanced fine-tuning-based method TFA. Furthermore, we interpret the experimental results of the baseline and find that due to the insufficient generalized and unique information of categories extracted from limited instances, the main error lies in the classification confusion of the classification branch. Thus, through introducing text-modal knowledge, a novel FSOD method for RSIs is proposed. It introduces a text knowledge extraction branch, including a text-modal knowledge extractor and a cross-modal assembly module. They extract generalized and unique characteristics from text-modal descriptions and fuse them into RoI visual features to highlight the corresponding information in visual-modal features, which effectively reduces classification confusion. Additionally, we propose a mask strategy and a separation loss for avoiding over-fitting and ambiguity of text-modal knowledge extraction. Extensive experiments on three datasets reveal the effectiveness and efficiency of our proposed method for both general and fine-grained object detection in RSIs under few-shot scenarios. In the future, our research will further investigate how to enrich the generalized information of categories in FSOD and explore more ways to automatically generate descriptions for categories to reduce the classification confusion of misclassifying novel classes into base classes or backgrounds.

APPENDIX
CLASS DESCRIPTIONS OF DATASETS

The text-modal description sentences of each category in the three datasets are presented in Table XII.

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